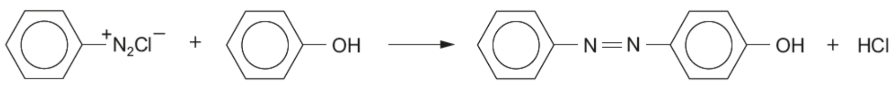
<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>10.</td><td>(a)</td><td></td><td></td><td>
<table>
<tr><th>Stage</th><th>Reagent used</th><th>Structural formula of product</th></tr>
<tr><td>1</td><td>KBr / H_2SO_4
or $SOCl_2$ / PCl_5</td><td>$CH_3CH_2CH_2Br$
$CH_3CH_2CH_2Cl$</td></tr>
<tr><td>2</td><td>KCN</td><td>$CH_3CH_2CH_2CN$</td></tr>
<tr><td>3</td><td>$LiAlH_4$</td><td></td></tr>
</table>
award (1) for each correct row</td><td>1</td><td>2</td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>sodium nitrate(III) / nitrite and acid e.g. HCl</td><td>1</td><td></td><td></td><td>1</td><td></td><td>1</td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>
accept other correct alternative equations</td><td></td><td>1</td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td></td><td>the statement is true as HCl will be removed as it is formed, moving the position of equilibrium to the right

accept other valid reasons</td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
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<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
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<tr><td></td><td>(d)</td><td></td><td></td><td>7.87 g of the amino acid gives 1470 cm^3 of N_2 gas (1)
$n(N_2) = 1470 / 24.5 \times 1000 = 0.060$
therefore 0.060 mol of the amino acid
molar mass of the amino acid = 7.87 / 0.060 = 131 (1)
$R—CH(NH_2)COOH \rightarrow 131$ and $CH(NH_2)COOH \rightarrow 74$
therefore $R \rightarrow 131 - 74 = 57$ (1)
it is a straight chain compound so R is $CH_3CH_2CH_2CH_2$
therefore amino acid must be $CH_3(CH_2)_3CH(NH_2)COOH$ (1)</td><td></td><td>2</td><td>2</td><td>4</td><td>2</td><td></td></tr>
<tr><td></td><td>(e)</td><td>(i)</td><td></td><td>2-methylpropane-1,3-dioic acid
accept 2-methylpropanedioic acid</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td><table><tr><th>Protons</th><th>Splitting pattern</th><th>Relative peak area</th></tr><tr><td>$-CH_2-CH_2-$</td><td>singlet / no splitting</td><td>4</td></tr><tr><td>$-C-H-$</td><td>quartet</td><td>1</td></tr><tr><td>$-CH_3$</td><td>doublet</td><td>3</td></tr></table>award (1) for each correct row</td><td></td><td>3</td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 10 total</td><td>3</td><td>8</td><td>3</td><td>14</td><td>2</td><td>1</td></tr>
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